

Lecture 7: Marginal & Joint Probability

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Topics of this lecture:

1. Random Variable
2. Joint Probability
3. Marginal Probability
4. Conditional Probability

1. Random Variable (X):

Random variable is a mathematical rule or a function that assigns a numerical value to the outcome of a random experiment.

Example: Random variable, $X = \text{Number appearing when I throw a die.}$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$P(X=1) = 1/6$$

$$P(X=2) = 1/6$$

$$P(X=3) = 1/6$$

$$P(X=4) = 1/6$$

$$P(X=5) = 1/6$$

$$P(X=6) = 1/6$$

Probability of a random variable: Probability of an event E of a random variable X is denoted as $P(X = E)$ or $P(E)$

For any random variable, sum of probability of all outcome in sample space is equal to 1.

If two random variables are dependent in that case there are three types of probability we need to consider those are:

1. Joint Probability
2. Marginal Probability
3. Conditional Probability

Random Experiment: Throwing a die and tossing a coin

X : Number appearing on a die; Possible values: $\{1, 2, 3, 4, 5, 6\}$

Y : Outcomes appearing on a toss; Possible values: $\{H, T\}$

2. Joint Probability:

Likelihood of two events happening simultaneously

$$P(A \cap B) \text{ or } P(X = A \text{ and } Y = B)$$

$$\begin{aligned} P(X = 3 \text{ and } Y = H) \\ &= P(X = 3 \cap Y = H) \\ &= 1/6 \times 1/2 \\ &= 1/12 \text{ (Though } X \text{ \& } Y \text{ are independent)} \end{aligned}$$

Sum of all possible values in a joint probability distribution is equal to 1.

Joint probability distribution table:

X/Y	X	Y
1	1/12	1/12
2	1/12	1/12
3	1/12	1/12
4	1/12	1/12
5	1/12	1/12
6	1/12	1/12

3. Marginal Probability:

Probability of a single event without considering the occurrence of other events.
It is obtained by summing the joint probabilities over the other events.

Marginal probability of getting a 3 on the die is

$$\begin{aligned} P(X = 3) &= P(X = 3, Y = H) + P(X = 3, Y = T) \\ &= 1/12 + 1/12 \\ &= 2/12 \\ &= 1/6 \end{aligned}$$

Marginal probability of getting a Head is

$$\begin{aligned} P(X = H) &= P(X = 1 \cap Y = H) + P(X = 2 \cap Y = H) + P(X = 3 \cap Y = H) + \dots + P(X = 6 \cap Y = H) \\ &= 1/12 + 1/12 + 1/12 + 1/12 + 1/12 + 1/12 \\ &= 6/12 \\ &= 1/2 \end{aligned}$$

X/Y	H	T
1	1/12	1/12
2	1/12	1/12
3	1/12	1/12
4	1/12	1/12
5	1/12	1/12
6	1/12	1/12

4. Conditional Probability:

probability of an event happening given that another event has already occurred.
It is denoted as $P(X | Y)$

Conditional probability of getting a Head given that the die shows at 3 is

$$P(Y = H | X = 3) = \frac{P(Y = H, X = 3)}{P(X = 3)}$$

$$= \frac{P(Y = H, X = 3)}{P(X = 3, Y = H) + (X = 3, Y = T)} = \frac{1/12}{1/6} = \frac{1}{2} = \frac{\text{Joint Probability}}{\text{Marginal Probability}}$$

Though tossing a coin is an independent event and its probability is always 1/2

Conditional Probability = Joint Probability / Marginal Probability